1.2

3. Which of the following formulas can be considered an algorithm for computing the area of a triangle whose side lengths are given positive numbers a, b, and c?

a. $S = \sqrt{p(p-a)(p-b)(p-c)}$, where $p = (a+b+c)/2$

b. $S = \frac{1}{2}bc \sin A$, where A is the angle between sides b and c

c. $S = \frac{1}{2}ah_a$, where h_a is the height to base a

a) $S = \sqrt{p(p-a)(p-b)(p-c)}$, $p = (a+b+c)/2$

If we assume that we know how to compute sqrt of an arbitrary value

This is an algorithm

b) $S = \frac{1}{2}bc \cdot \sin A$ → A → how to compute A

→ sine → how to compute sine of an angle

NOT an algorithm

c) $S = \frac{1}{2}a h_a$ → how to compute ?

NOT an algorithm

1.2

4. Write pseudocode for an algorithm for finding real roots of equation $ax^2 + bx + c = 0$ for arbitrary real coefficients a, b, and c. (You may assume the availability of the square root function $sqrt(x)$.)

$ax^2 + bx + c = 0$

$r_1, r_2 = \frac{-b \mp \sqrt{\Delta}}{2a}$, $\Delta = b^2 - 4ac$

If $\Delta > 0$, real roots

If $\Delta = 0$, one root

If $\Delta < 0$, 'no real roots'

Algorithm A(a, b, c)

// Input: Real coefficients a, b, c

// Output: The real roots of given equation $ax^2+bx+c=0$

or return a message about absence of roots

If $a \neq 0$

$D \leftarrow b*b - 4*a*c$ // $\Delta = b^2 - 4ac$

If $D > 0$

temp $\leftarrow 2*a$

$r_1 \leftarrow (-b + sqrt(D)) / temp$

$r_2 \leftarrow (-b - sqrt(D)) / temp$

return r_1, r_2

elif D = 0

return r ← -b/(2*a)

else // D < 0

return 'no real roots'

else // a = 0

if b ≠ 0

return -c/b

else // a = 0 , b = 0

if // c = 0

return 'All real roots'

else // c ≠ 0

return 'no real roots'

1.2

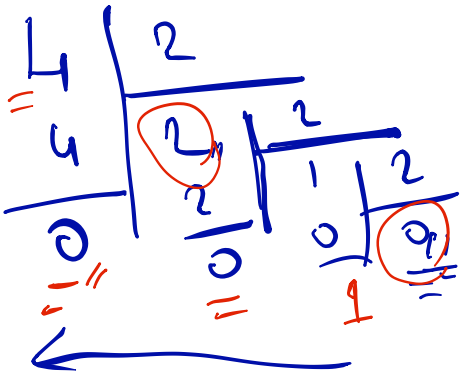
5. Describe the standard algorithm for finding the binary representation of a positive decimal integer

a. in English.

b. in pseudocode.

$(4)_2 = 100$

$n = 4$



a) 1) Divide the given number n by 2

2) The remainder (0 or 1) will be the next (right to left) binary digit

3) Replace n by the quotient of the last division and repeat this operation until n becomes 0.

b) Algorithm Binary (n)

// input : A positive integer number n

// output : $b_k, b_{k-1}, b_{k-2}, \ldots, b_0$ of n's digits

while $n \neq 0$

$b_k \leftarrow n \bmod 2$ → 2

$n \leftarrow \lfloor n/2 \rfloor$ → 1, 3

$k \leftarrow k+1$

return all b_k's in decreasing order

$n = 4$

$k \leftarrow 0$

while $4 \neq 0$

$b_0 = 0$

$n = 2$

$k \leftarrow 1$

$2 \neq 0$

$b_1 = 0$

$n = 1$

$k = 2$

$1 \neq 0$

$b_2 = 1$

$n = 0$

$k = 3$

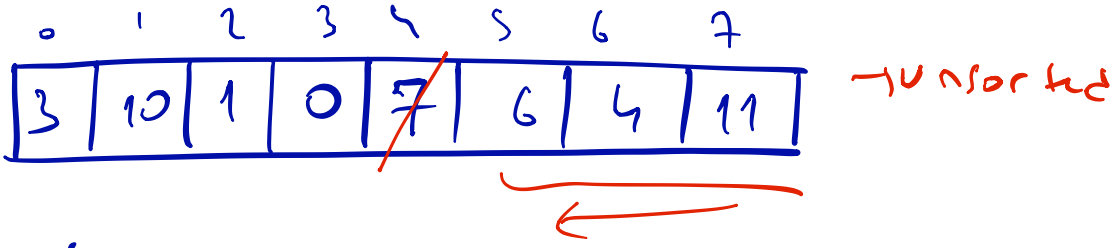
$0 = 0$ ✗

$b_2\ b_1\ b_0 \rightarrow 100$

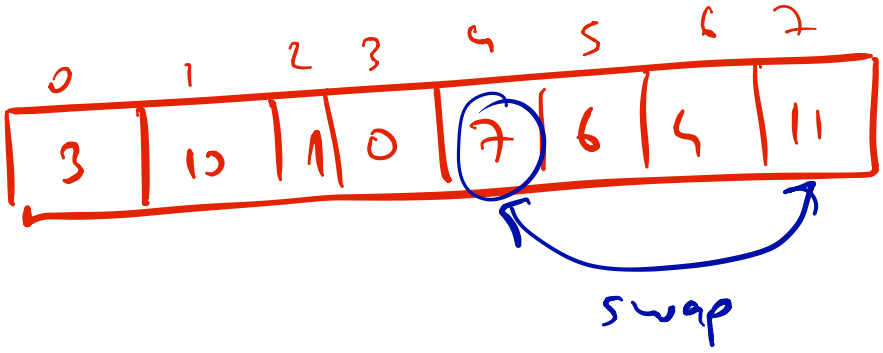
1.4

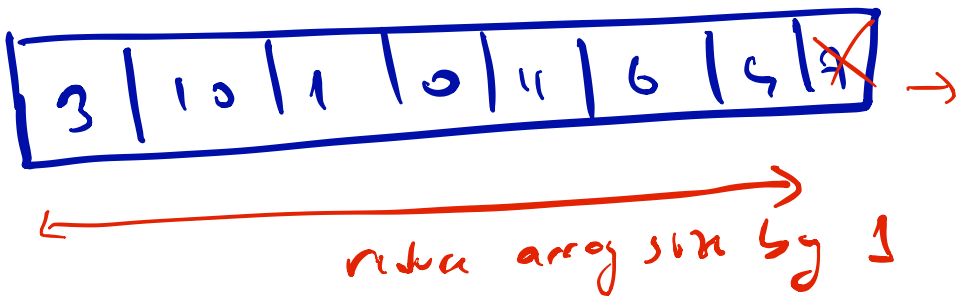
1. Describe how one can implement each of the following operations on an array so that the time it takes does not depend on the array's size n.
   - **a.** Delete the ith element of an array ($1 \le i \le n$).
   - **b.** Delete the ith element of a sorted array (the remaining array has to stay sorted, of course).

a)

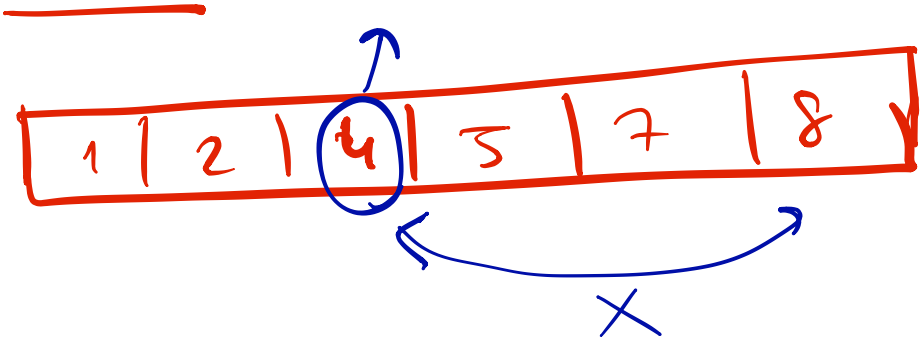


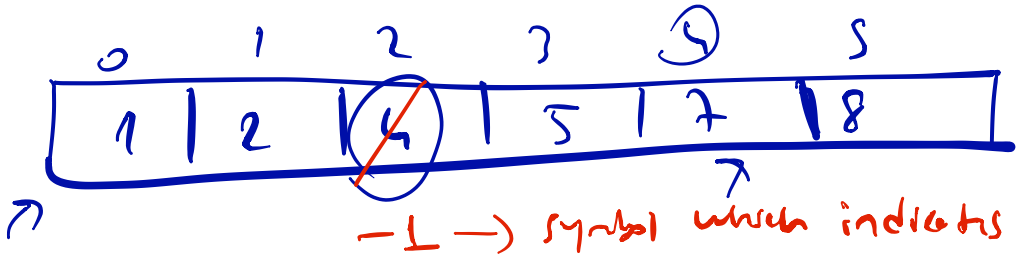
$i = 4$





b) Sorted

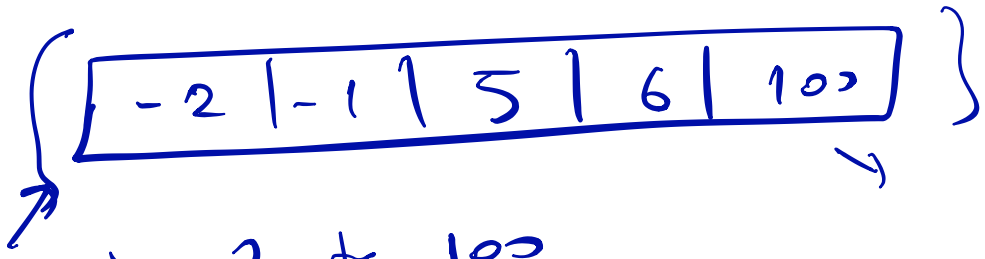




A[4]=2

-1 → symbol which indicates the cell empty

A[2] = -1



→ -2 + 100

→ -3 , 101 , -10000 ---

symbols

2.1

9. For each of the following pairs of functions, indicate whether the first function of each of the following pairs has a lower, same, or higher order of growth (to within a constant multiple) than the second function.

a. $n(n+1)$ and $2000n^2$

b. $100n^2$ and $0.01n^3$

c. $\log_2 n$ and $\ln n$

d. $\log_2^2 n$ and $\log_2 n^2$

e. 2^{n-1} and 2^n

f. $(n-1)!$ and $n!$

a) $n(n+1) = n^2 + n \approx n^2$ → Informal

$2000 \cdot n^2$ ✓ same order

→ 1st function has same order of growth as 2nd func.

Formal way

$$\lim_{n\to\infty} \frac{n(n+1)}{2000n^2} = \lim_{n\to\infty} \frac{n^2+n}{2000n^2} = C$$

→ If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = C$, $f(n)$ has same order of growth with $g(n)$

b) $100n^2 > 0.01n^3$

1st function has lower order of growth than 2nd func.

→ $\lim_{n\to\infty} \frac{100 \cdot n^2}{0.01n^3} = 0$ →

If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0$, $f(n)$ has lower order of growth than $g(n)$

c) $\log_2 n \to \log_2 n = \frac{\log n}{\log 2}$

$\ln n \to \ln n = \frac{\log n}{\log e}$

$$\lim_{n\to\infty} \frac{\log_2 n}{\ln n} = \frac{\frac{\log n}{\log 2}}{\frac{\log n}{\log e}} = \frac{\log e}{\log 2} = c$$

1st funct. has same order of growth as 2nd funct.

d) $\log^2 2^n = (\log_2 n)^2 = \log_2 n \cdot \log_2 n$

$\log_2 n^2 = 2 \log_2 n$

$$\lim_{n\to\infty} \frac{\log_2^2 n}{\log_2 n^2} = \frac{\log_2 n \cdot \log_2 n}{2 \log_2 n} = \frac{\log_2 n}{2} = \infty$$

If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty$, $f(n)$ has higher order of growth than $g(n)$

e) $2^{n-1} = \frac{2^n}{2}$

2^n ✓

$$\lim_{n\to\infty} \frac{\frac{1}{2}\cdot 2^n}{2^n} = \frac{1}{2}$$

These two functions have same order of growth.

f) $(n-1)!$

$n!$ → $n! = n\cdot(n-1)!$

$$\lim_{n\to\infty} \frac{(n-1)!}{n!} = \frac{(n-1)!}{n\cdot(n-1)!} = \frac{1}{n} = 0$$

1st funct has lower order of growth than 2nd func.

2.1

2. Use the informal definitions of O, Θ, and Ω to determine whether the following assertions are true or false.

a. $n(n+1)/2 \in O(n^3)$ **b.** $n(n+1)/2 \in O(n^2)$

c. $n(n+1)/2 \in \Theta(n^3)$ **d.** $n(n+1)/2 \in \Omega(n)$

a) $\frac{n(n+1)}{2} \in O(n^3)$ TRUE

$\rightarrow \lim_{n\to\infty} \frac{\frac{n(n+1)}{2}}{n^3} = \frac{n^2+n}{2n^3} = 0$

$\rightarrow$ $T(n)$ is $O(f(n))$ means that $f(n)$ describe the upper bound for $T(n)$

b) $n(n+1) \in O(n^2)$ TRUE

$\rightarrow$ $T(n)$ is $O(f(n))$ iff $T(n) \leq c \cdot f(n)$

$n(n+1) \approx n^2 \in O(n^2)$

c) $\frac{n(n+1)}{2} \in \Theta(n^3)$ FALSE

d) $\frac{n(n+1)}{2} \in \Omega(n)$ TRUE

$\frac{n(n+1)}{2}$ grows at least as fast as n.

2.2

3. For each of the following functions, indicate the class $\Theta(g(n))$ the function belongs to. (Use the simplest $g(n)$ possible in your answers.) Prove your assertions.

a. $(n^2+1)^{10}$

b. $\sqrt{10n^2+7n+3}$

c. $2n\lg(n+2)^2 + (n+2)^2 \lg \frac{n}{2}$

d. $2^{n+1} + 3^{n-1}$

e. $\lfloor \log_2 n \rfloor$

a) $(n^2+1)^{10} \approx n^{20} \in \Theta(n^{20})$

b) $10n^2+7n+3 \approx 10n^2$

$\sqrt{10n^2} = n\sqrt{10} \in \Theta(n)$

c) $2n \log (n+2)^2 + (n+2)^2 \lg \frac{n}{2}$

$\approx n^2$

$2 \cdot 2 n \lg(n+2) + n^2 \lg \frac{n}{2}$

$\Theta(n \lg n) + \Theta(n^2 \lg n)$

$= \Theta(n^2 \lg n)$

d) $2^{n+1} + 3^{n-1} = 2 \cdot 2^n + \frac{3^n}{3}$

$\Theta(2^n) + \Theta(3^n)$

$= \Theta(3^n)$

e) $\lfloor \log_2 n \rfloor \in \Theta(\log_2 n)$

→ inequality

$\lfloor 4.9 \rfloor = 4$

$4 \leq 4$

→ $x - 1 < \lfloor x \rfloor \leq x$

↳ use

$\log_2 n - 1 < \lfloor \log_2 n \rfloor \leq \log_2 n$

$\lfloor \log_2 n \rfloor \in \Theta(\log_2 n)$

2.2

5. List the following functions according to their order of growth from the lowest to the highest:

$$(n-2)!,\quad 5\lg(n+100)^{10},\quad 2^{2n},\quad 0.001n^4+3n^3+1,\quad \ln^2 n,\quad \sqrt[3]{n},\quad 3^n.$$

a $(n-2)! \rightarrow \in \Theta((n-2)!)$

b $5\log(n+100)^{10} = 50\log(n+100) \in \Theta(\lg n)$

c $2^{2n} \rightarrow 4^n \in \Theta(4^n)$

d $0.001n^4 + 3n^3 + 1 \approx n^4 \in \Theta(n^4)$

e $\ln^2 n = \ln n \cdot \ln n \rightarrow \in \Theta(\log^2 n)$

f $\sqrt[3]{n} = n^{1/3} \rightarrow \in \Theta(n^{1/3})$

g $3^n \rightarrow \in \Theta(3^n)$

b, e, f, d, g, c, a → from lowest to highest

→ constant, logarithmic, linear, nlgn, quadratic, cubic, exponential, factorial n^n

2.3

2. Find the order of growth of the following sums. Use the $\Theta(g(n))$ notation with the simplest function $g(n)$ possible.

a. $\sum_{i=0}^{n-1}(i^2+1)^2$ **b.** $\sum_{i=2}^{n-1} \lg i^2$

c. $\sum_{i=1}^{n}(i+1)2^{i-1}$ **d.** $\sum_{i=0}^{n-1}\sum_{j=0}^{i-1}(i+j)$

a) $\sum_{i=0}^{n-1}(i^2+1)^2 = \sum_{i=0}^{n-1}(i^4+2i^2+1)$

$= \sum_{i=0}^{n-1} i^4 + 2\cdot\sum_{i=0}^{n-1} i^2 + \sum_{i=0}^{n-1} 1$

prof

↳ $\sum_{i=0}^{n-1} i^4 = \frac{n(n+1)(6n^3+9n^2+n-1)}{30} \approx n^5$

$\sum_{i=0}^{n} \frac{n(n+1)(2n+1)}{6} \Theta(n^5)$

↖ $2\sum_{i=0}^{n-1} i^2 = \frac{(n-1)(n-1+1)(2(n-1)+1)}{6} \approx n^3$

$\Theta(n^3)$

$\sum_{i=0}^{n-1} 1 = 1\cdot(n-1-0+1) = n$

$\Theta(n)$

$\Theta(n^5) + \Theta(n^3) + \Theta(n) = \Theta(n^5)$

b) $\sum_{i=2}^{n-1} \log_2 i^2 = \sum_{i=2}^{n-1} 2\log_2 i = 2\sum_{i=2}^{n-1} \log_2 i$

$= 2\sum_{i=1}^{n} \log_2 i - 2\sum_{i=1}^{1} \log_2 i - 2\sum_{i=n}^{n} \log_2 i$

$= 2\sum_{i=1}^{n} \log_2 i - 0 - 2\log_2 n$

$= 2\sum_{i=2}^{n-1} \log_2 i$

$= 2\sum_{i=1}^{n} \log_2 i - 2\log_2 n$

$2\sum_{i=1}^{n} \log_2 i \rightarrow \Theta(n \log_2 n)$ (n terms)

$2\log_2 n \rightarrow \Theta(\log_2 n)$

$2(\log_2 1 + \log_2 2 + \cdots + \log_2 n) \leq \log_2 n + \cdots \log_2 n$

$2\log_2 (1 \cdot 2 \cdots n)$

$= 2\log_2 n! \leq n \log_2 n$

$\Rightarrow \Theta(n \log_2 n)$

$\rightarrow \Theta(n \log_2 n) + \Theta(\log_2 n) = \Theta(n \log_2 n)$

c) $\sum_{i=1}^{n} (i+1)\, 2^{i-1} = \sum_{i=1}^{n} \left(2^{i-1} \cdot i + 2^{i-1} \right)$

$\sum_{i=1}^{n} 2^{i-1} \cdot i + \sum_{i=1}^{n} 2^{i-1}$

Fórmula

$$\sum_{k=0}^{n} a^k = \frac{a^{n+1} - 1}{a - 1}$$

$$S = \sum_{i=1}^{n} i \cdot 2^{i-1} = 1 \cdot 2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \dots + n \cdot 2^{n-1}$$

$$S = 2S - S = 2\left(2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \dots + n \cdot 2^{n-1}\right) - \left(2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \dots + n \cdot 2^{n-1}\right)$$

$$= 2 \cdot 2^0 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + (n-1) \cdot 2^{n-1} + n \cdot 2^n - 2^0 - 2 \cdot 2^1 - 3 \cdot 2^2 - 4 \cdot 2^3 - \dots - n \cdot 2^{n-1}$$

$$= -1 - 2^1 - 2^2 - 2^3 - \dots - 2^{(n-1)} + n \cdot 2^n$$

$$-1\left(1 + 2^1 + 2^2 + \dots + 2^{(n-1)}\right)$$

$$= -\frac{2^{n-1+1} - 1}{2 - 1} + n \cdot 2^n$$

$$= 1 - 2^n + n \cdot 2^n = 2^n(n-1) + 1$$

$$\Theta(n \cdot 2^n)$$

$$\sum_{i=1}^{n} 2^{i-1} = \frac{2^{n-1+1} - 1}{2-1} = 2^n - 1 \rightarrow \Theta(2^n)$$

$$\Rightarrow \Theta(n \cdot 2^n) + \Theta(2^n) = \Theta(2^n \cdot n)$$